

CII SR -15th Edition Online Kaizen Compétition 2020

A Kaizen Presentation By

(Everest Industries Ltd)

Podanur Works - Coimbatore

18th ~ 20th August 2020

COMPANY OVERVIEW

Everest Industries Limited, incorporated in 1934, has a rich history in manufacturing of Building products and Steel products. Everest offers a complete range of roofing ceiling, wall, flooring and cladding products distributed through a large network, and also pre-engineered steel buildings for industrial, commercial and residential applications. It is one of the leading building solution providers in India, providing detailed technical assistance in the form of design, drawings and implementation for every project.

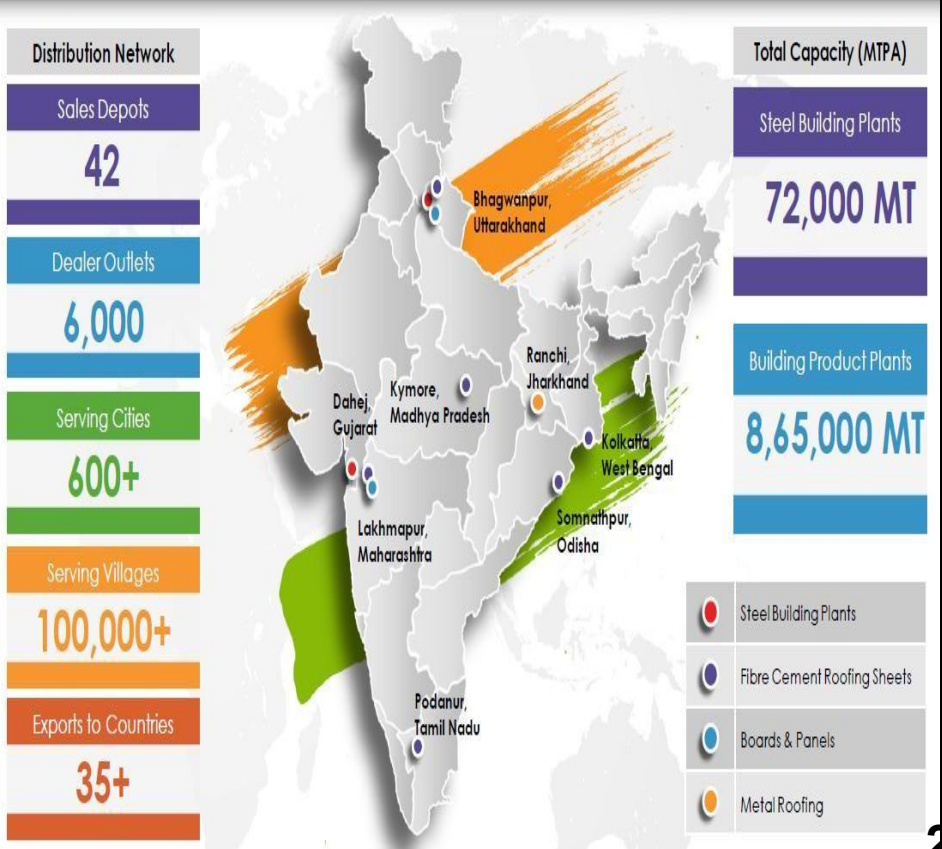
OUR BUSINESS

The company businesses are divided into two distinct business segments with capacity –

- 1. Buildings Products – 8,65,000 MT / Annum
- 2. Steel Buildings – 72,000 MT / Annum

Buildings Products segment contributed 63% of the company’s revenue and includes Fiber cement roofing sheets, Fiber cement boards, Rapicon wall & their accessories.

Steel Building segment accounted for 37% of annual revenue. In this segment Company offers customized building solutions in steel such as Pre-Engineered steel Buildings, Smart steel buildings, Metal roofing and Cladding.



Company Products & Customers



Products:



Fibre Cement Roofing Sheets



Hi-Tech Roofing Sheets



Super Sheets



Cement Planks



Fibre Cement Boards



Rapicon Walls



Smart Steel Buildings



Customers:



Reliance
Industries Limited



Simplex Infrastructures Limited



Kaizen Theme: To improve productivity & to reduce maintenance cost of sieve filtration system.

Implemented Area: Shopfloor, Podanur Works, Coimbatore

Implementation Date: 16th November 2019

Team:



Mr. P. Ganesan
(Sieve Fitter)

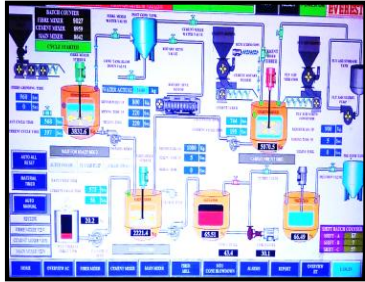


**Mr. Prince
Mariyanayagam**
(Sieve Fitter)

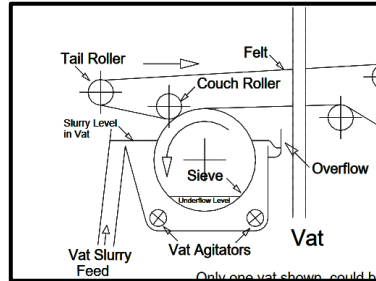


Mr. K. Manikam
(Sieve Fitter)

Process Flow:



1) RAW MATERIAL PREPARATION



2) VATS



3) SHEET FORMING



4) SHEET HANDLING/
PILING MACHINE



6) DE-PILING MACHINE



5) PRE-CURING
TUNNEL

Problem/ Present Status :

Frequency of sieve failure is high. By this, lifetime of the sieve is less & machine breakdown is high.

SIEVE DESCRIPTION



SIEVE FRAME



INNER MESH

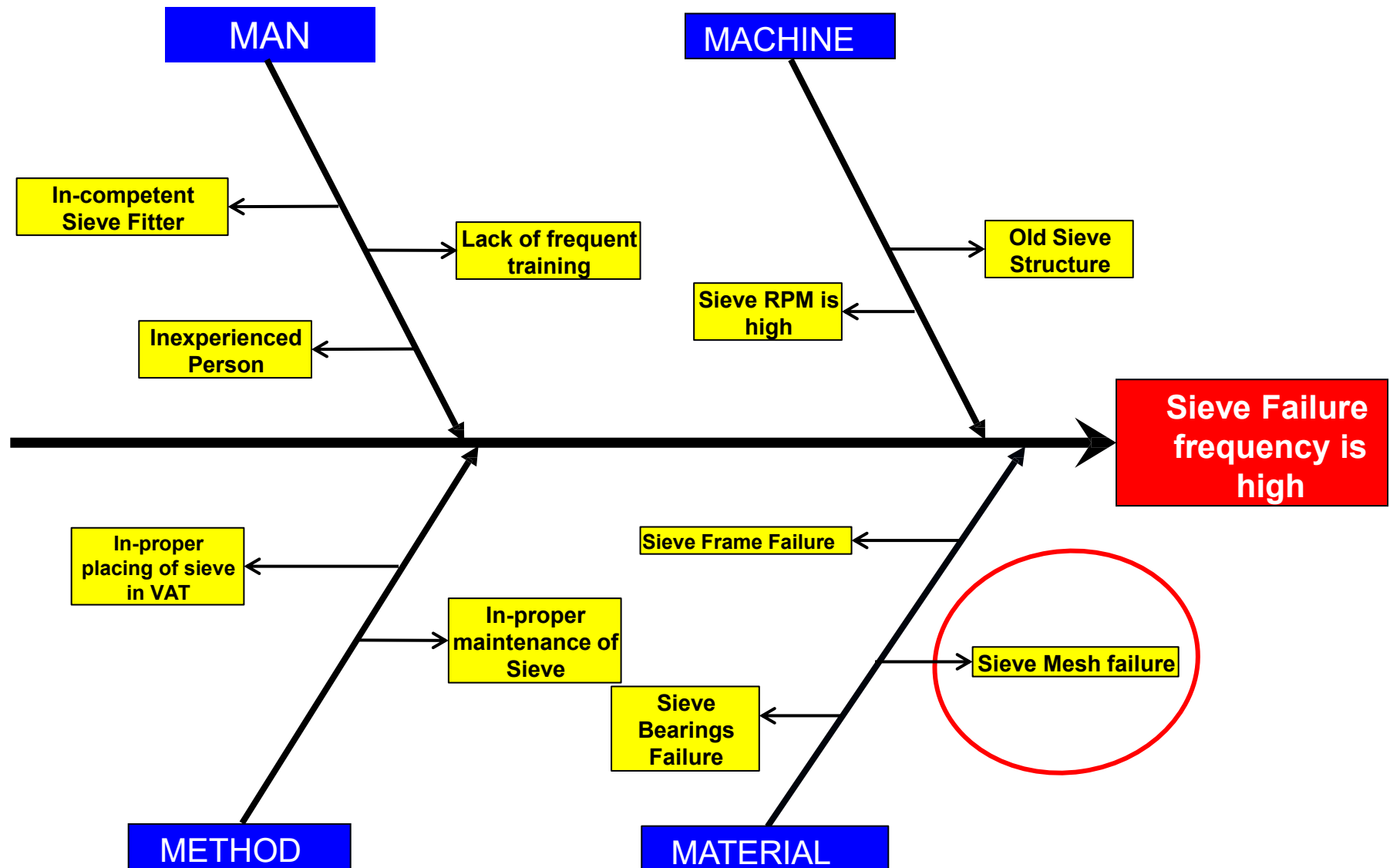


OUTER MESH

Problem/ Present Status :

Frequency of sieve failure is high. By this, lifetime of the sieve is less & machine breakdown is high.

Root Cause - Cause & Effect Analysis



Root Cause - Why Why Analysis



Problem Statement	Sieve Mesh Failure
Why 1	More load on the outer mesh.
Why 2	When sieve is operational, inner mesh tightness is loosening & becoming uneven for supporting outer mesh.
Why 3	Inner mesh ends are disjointed.
Why 4	Inner mesh ends are joined with metal ties.

Root Cause: Inner mesh ends are joined with metal ties.

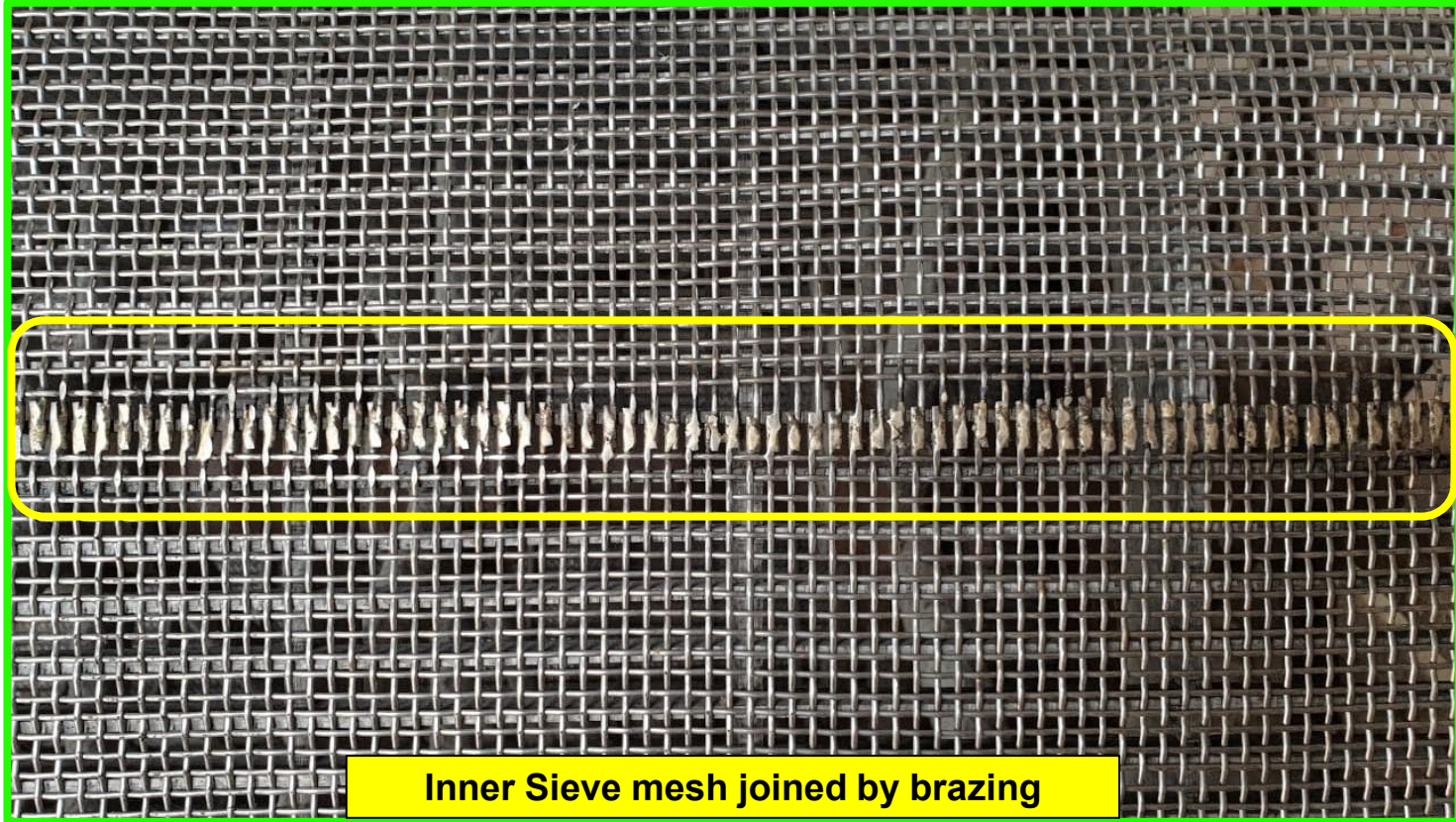
Idea Generation: To join inner mesh ends by brazing.

Action: To join inner mesh by brazing. By this there will not be loosening of sieve inner mesh during operational. And there won't be any damages to outer sieve mesh.

S.No	Counter Measure Activities	Start Date	Target Date	Responsibility	Status
01	To study causes of sieve failure.	08/10/2019	13/10/2019	Sieve Section & Production Department	Completed
02	To study & take trails on suitable brazing for joining inner mesh ends.	14/10/2019	20/10/2019	Sieve Fitters	Completed
03	To remove metal ties of inner mesh & to join the mesh by brazing.	21/10/2019	26/11/2019	Sieve Fitters	Completed



- ❖ Previously, inner sieve mesh ends are joined together with metal ties. By this, when sieve is in operational, metal ties will become loose & inner sieve will become uneven. Due to this, outer mesh is experiencing more load. This leads to frequent sieve failures.



- ❖ Now, inner sieve mesh is brazed instead of joining with metal ties. By this, sieve mesh became very rigid & sieve lifetime has increased by 150%.



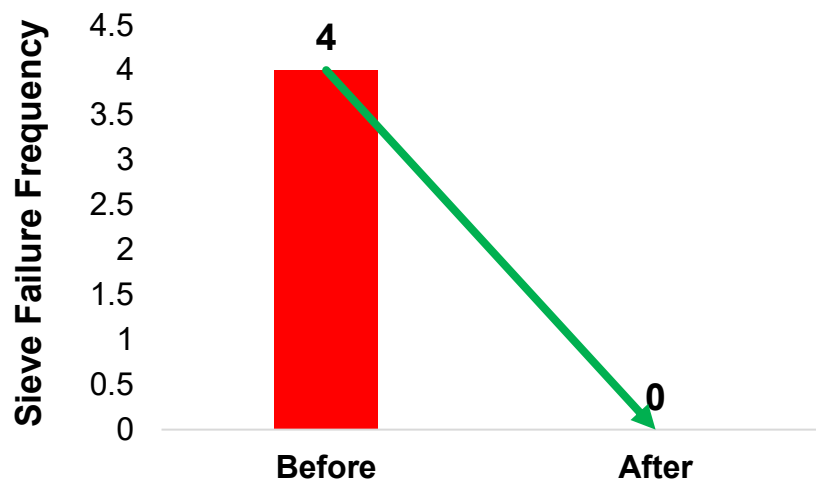
Inner Sieve mesh joined by Brazing

- ❖ Now, inner sieve mesh is brazed instead of joining with metal ties. By this, sieve mesh became very rigid & sieve lifetime has increased by 150%.

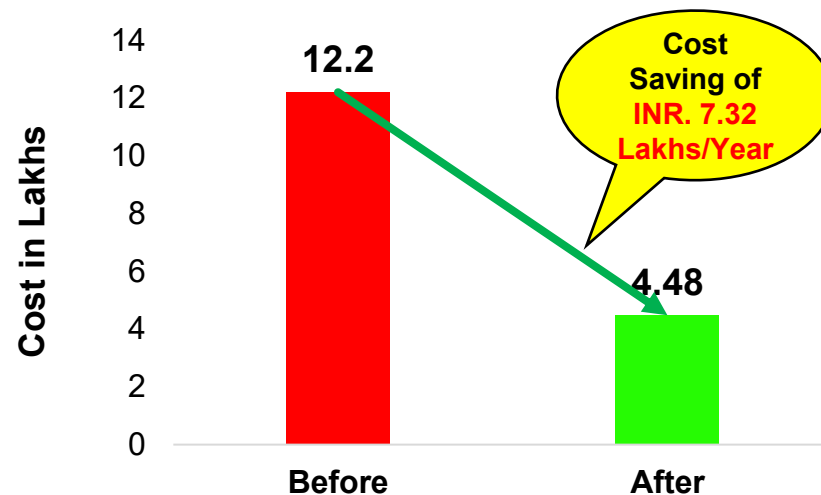
Sieve Failure & Cost Consumption Data

	Before	After
Sieve Failure Frequency / Month	4	0
Sieve Operational Time in Hours	1000 Hours	2500
Sieve Operational Days	41	104
Maintenance Cost of 1 Sieve mesh	INR. 13,193/- for 41 Days	IN. 13,193/- for 104 Days
Cost Per Year for 1 Sieve mesh	INR. 1,01,702/-	INR. 40,680/-
For 12 Sieve meshes Cost/Year	INR. 12,20,426/-	INR. 4,48,171/-
Cost Saving / Year	INR. 7,32,255/-	

Sieve Failure Frequency/Month



Cost of Sieve Mesh / Year



Tangible Benefits

- Lifetime of the Sieve has increased by 150%.
- Sieve mesh cost is reduced by 60%.
- Cost Saving of INR. 7.32 Lakhs/Year.
- Sieve failure during operational has been eliminated.
- Breakdown hours reduced.
- Revenue loss reduced.

In-tangible Benefits

- Morale of our People increased due to increase in Productivity & less maintenance of sieves.
- Safety of people increased by avoiding multiple sieve handling.

- Training on Updated Work instructions are given to all the workmen for regular inspection & maintenance.

	EVEREST INDUSTRIES LIMITED PODANUR WORKS IS/ISO 9001 : 2015 WORK INSTRUCTION	DOC NO : EL/WI/603 REV NO : 01 PAGE : 1 OF 1 DATE : 29/11/2019
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Work Instruction: WI 603
MAINTENANCE OF SIEVES

Procedure:

1. Both side bearing should be checked and greased.
2. Both side border ring should be checked / replaced if required.
3. Remove old meshes carefully from the sieve.
4. Spread new inner mesh and tighten it by using mesh holders.
5. Join inner Sieve mesh by welding
6. Spread outer sieve mesh on inner mesh & tighten it by using mesh holders.
7. Stitch outer mesh end.

Kaizen Sustenance:

WHAT TO DO	Regular Inspection & Maintenance.
HOW TO DO	Regular Training on Standardized work instructions
FREQUENCY	Weekly

Horizontal Deployment:

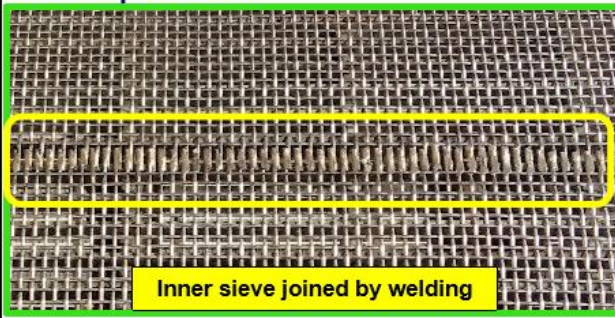
BW	CW	KW	LW	PW	SW



- Under Progress



- Deployed

Kaizen Sheet							Company	MM/YY	S.No												
Productivity	Quality	Delivery	Safety	Material	Energy	Maintenance	Everest Industries Ltd	11/19	1												
Kaizen Title: To improve productivity & to reduce maintenance cost of sieve filtration system.							Implemented Area: Sieve Section														
Problem/Present Status: Frequency of sieve mesh failure is high. By this, Lifetime of the Sieve is less & machine breakdown is high.			Before Improvement:  Inner sieve joined with metal ties				Implemented by: P. Ganesan, Prince Marianayagam, K. Manikam (Sieve Section)														
Root Cause Identification:			After Improvement:  Inner sieve joined by welding				Result/Benefit:														
Why 1: More load on the outer mesh.							(a) Qualitative Lifetime of the sieve has increased by 150%.														
Why 2: When sieve is operational, inner mesh tightness is loosening & becoming uneven to support outermesh.							(b) Quantitative														
Why 3: Inner mesh ends are disjointed.							- Sieve mesh cost is reduced by 63%. - Cost Saving of INR. 7.32 Lakhs/Year														
Why 4: Innermesh ends are joined with metal ties.							 <table border="1"> <caption>Cost of Sieve Mesh / Year</caption> <thead> <tr> <th>Period</th> <th>Cost in Lakhs</th> </tr> </thead> <tbody> <tr> <td>Before</td> <td>12.2</td> </tr> <tr> <td>After</td> <td>4.48</td> </tr> </tbody> </table>			Period	Cost in Lakhs	Before	12.2	After	4.48						
Period	Cost in Lakhs																				
Before	12.2																				
After	4.48																				
Root cause			Inner mesh ends are joined with metal ties.				Standardization:														
Idea to eliminate root cause			To braze inner mesh ends.				Training is provided on SOP to workmen.														
Counter-measure			Inner mesh ends are joined by brazing. By this, inner mesh will be in even condition during operation & there won't be any damages to outer sieve mesh.				How many places this Kaizen can be deployed horizontally														
							<table border="1"> <thead> <tr> <th>BW</th> <th>CW</th> <th>KW</th> <th>LW</th> <th>PW</th> <th>SW</th> </tr> </thead> <tbody> <tr> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> </tr> </tbody> </table>			BW	CW	KW	LW	PW	SW						
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Thank You